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Human Brain Mapping
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Take-Home Assignment-3

This assignment report exhibits an analysis of ERPs or event-related potentials from a single-subject auditory oddball task via the OpenNeuro dataset (ds003061). A 64-channel BioSemi EEG dataset was preprocessed by utilizing MNE-Python, along with a bad-channel interpolation, bandpass filtering (0.1–30 Hz), 50 Hz notch-filtering, ICA-based ocular artifact removal and average re-referencing. Epochs were also extracted from standard (500 Hz, 70%), oddball target (15%), and noise distractor (15%) stimuli. For this, two ERP components were analyzed i.e., a) the P300 denoting a positive deflection at 300-500 ms over parietal scalp which was reportedly significantly larger for oddball as opposed to the standard stimuli, overall consistent with the context-updating theory, and b) the N200 representing a negative deflection at 180–280 ms over frontocentral scalp which was also evidently more negative for the oddball stimuli, further indicating the pre-attentive deviance detection. MMN or the mismatch negativity difference wave moreover confirmed an automatic auditory change detection. Furthermore, four referencing schemes i.e., average, Cz, linked mastoids, and Fpz were contrasted resulting in a demonstration of how reference choice repositions topographic distributions. Therefore, the average reference here itself provided perhaps the clearest mapping of the P300 generator.

I. Part-I: Data Cleaning

A. Data Loading & Inspection: The raw EEG data files were loaded from the EEGLAB .set file (sub-001_task-P300_run-1_eeg.set) via the mne.io.read_raw_eeglab() function with preload set as True. The dataset was recorded by utilizing a 64-channel BioSemi Active-2 system at a sampling rate of 256 Hz, with a total recording duration of approximately 13 minutes. Channel types were corrected in accordance with the BIDS channels.tsv metadata by reclassifying non-EEG channels (i.e., GSR, respiration, temperature) as miscellaneous. Also, events were loaded from the BIDS events.tsv file and further mapped to integer codes: standard = 1 (522 trials, 500 Hz tone), oddball = 2 (113 trials, combining oddball_with_reponse and oddball labels), and noise = 3 (112 trials, combining noise and noise_with_reponse labels). Additionally, button-press response and ignore-events were excluded from epoching. Then, the electrode montage was assigned using Cartesian coordinates from the electrodes.tsv file which was converted from millimeters to meters and later applied as a custom MNE montage in the very head coordinate frame.

B. Bad-Channel Detection & Interpolation: Bad-channels were carefully identified via an automated log-variance outlier detection method. The log-transformed variance of each channel's time was computed so that channels with z-scores exceeding 3.0 from the median log-variance could be flagged as pathologically noisy or simply flat. Here, a safety cap of 10% of the total channels prevented over-rejection. Therefore, the identified bad-channels were interpolated via spherical spline interpolation mechanism (MNE's interpolate_bads() function).

C. Filtering: FIR or a finite impulse response bandpass filter with a Hann-window was applied between 0.1 and 30 Hz. The 0.1 Hz high-pass cutoff removed slow drifts and DC offsets, whilst the 30 Hz low-pass further attenuated a high-frequency muscle (EMG) artifacts and later

preserved all ERP components of interest (P300, N200, MMN). Hence, a notch filter at 50 Hz was further applied to ensure the removal of power-line noise, as the recording was primarily conducted in India where the mains frequency resides at approximately 50 Hz.

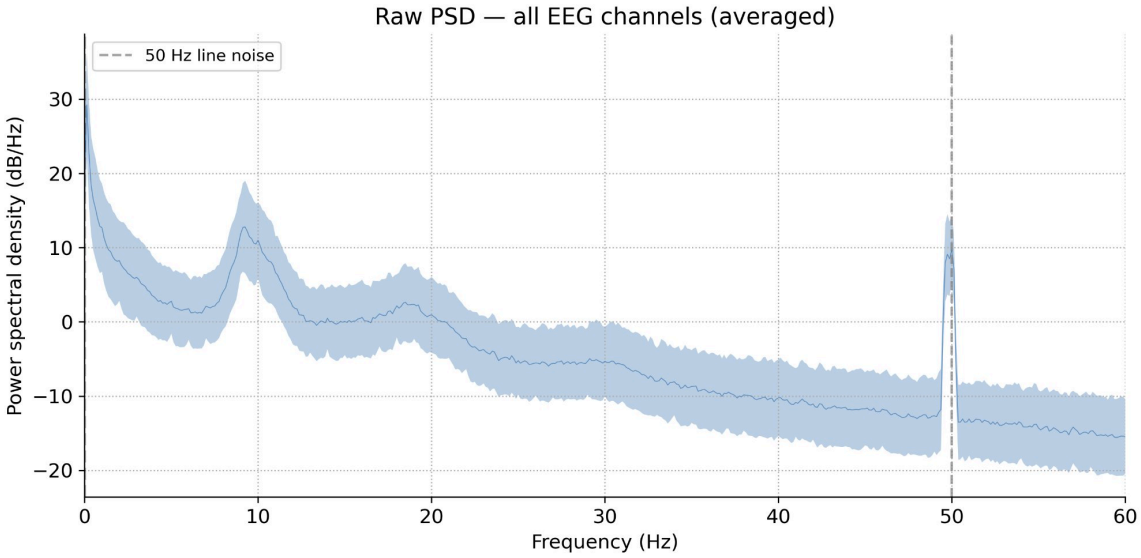


Figure-1: A power-spectral density of raw EEG data across all 64 channels. The prominent peak at 50 Hz indicates the power-line noise. Here, a clear alpha peak is visible around 8-12 Hz.

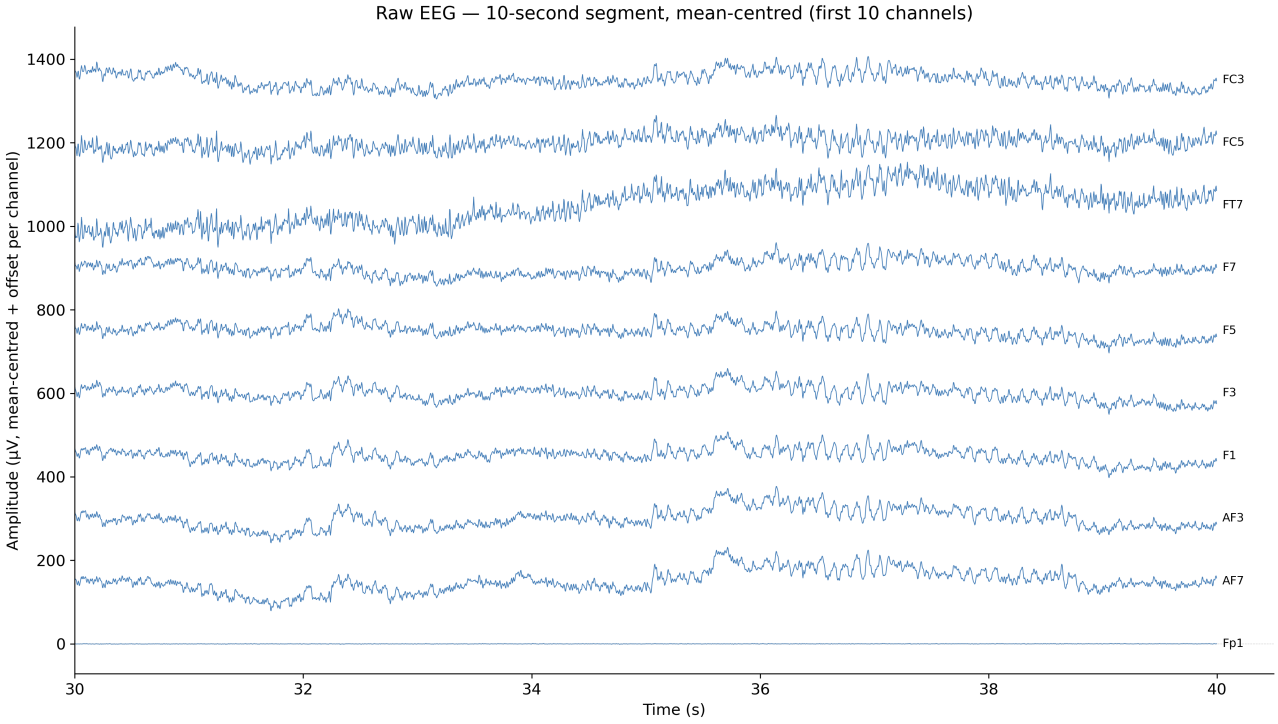


Figure-2: A raw EEG time-series for the first 10 channels (30-40s segment), indicating a large amplitude variability as well as the potential artifacts.

D. ICA-Artifact Removal: An Independent Component Analysis or ICA was conducted via the FastICA algorithm comprising 15 components and fixed random seed (i.e., 42) for

reproducibility. Since no dedicated EOG channel was recorded, the channels Fp1 and Fp2 were used as the frontal EOG proxies. Hence, each IC time-course was correlated via the Pearson r with the Fp1 and Fp2 signals and components with $|r| > 0.3$ were flagged as ocular artifacts and therefore were excluded. The ICA topographic maps of all 15 components are shown below as Figure-3.

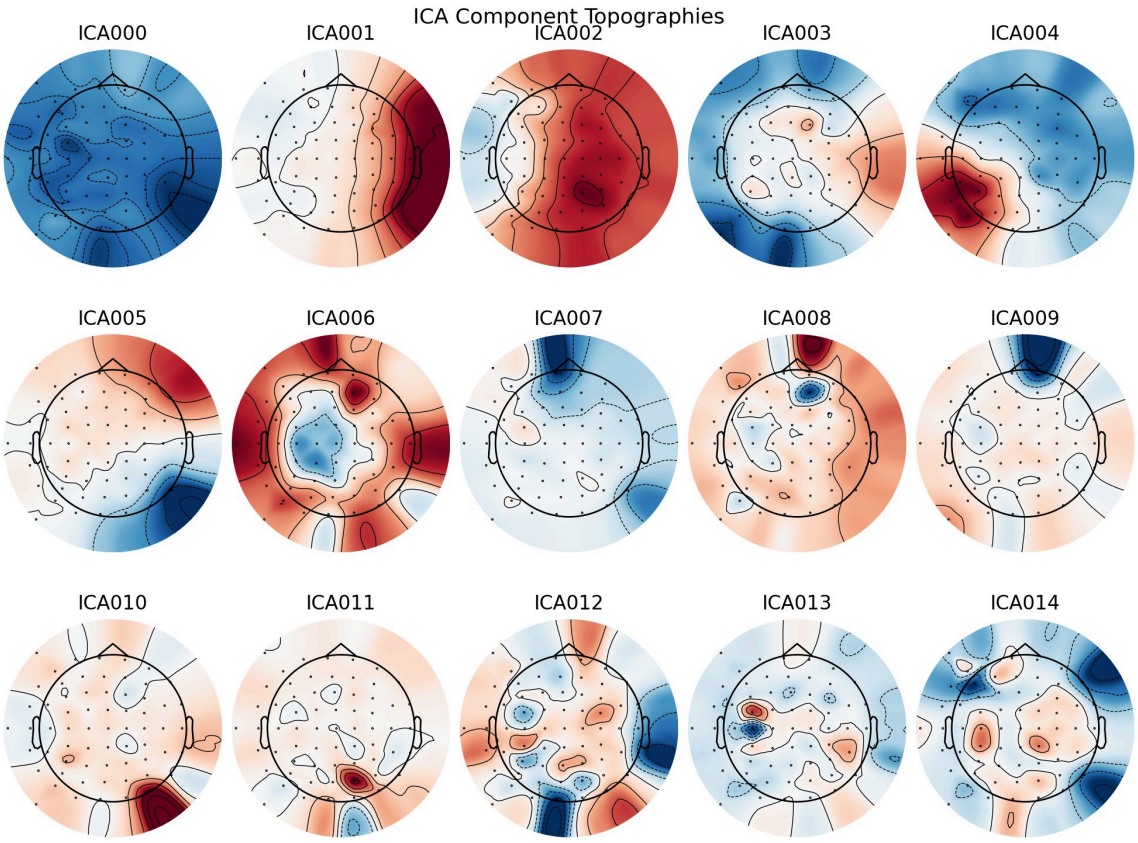


Figure-3: The topographic maps of all 15 ICA components. Here, the components indicating frontal polarity patterns were identified via correlation with FP1 and Fp2, and hence, excluded.

E. Re-referencing: The data were re-referenced to the average of all the EEG channels i.e., the average reference is optimal when electrode coverage is approximately uniform over the scalp, as in this 64-channel BioSemi cap. Hence, under this condition the sum of all electrode potentials approaches zero, thereby providing a distribution that ranges close to the true reference-free potential (Dien, 1998; Luck, 2014).

F. Preprocessing Comparison: The Figure-4 below presents the post-preprocessing PSD highlighting the successful removal of the 50 Hz line noise peak and the sharp rolloff that occurs above 30 Hz from the bandpass filter. Moreover, Figure-5 further exhibits the cleaned time-series, where ocular artifacts have been visibly attenuated while the cortical alpha rhythms (8-12 Hz) are more clearly resolved. In addition, Figure-6 further shows a direct side-by-side comparison of raw and preprocessed data.

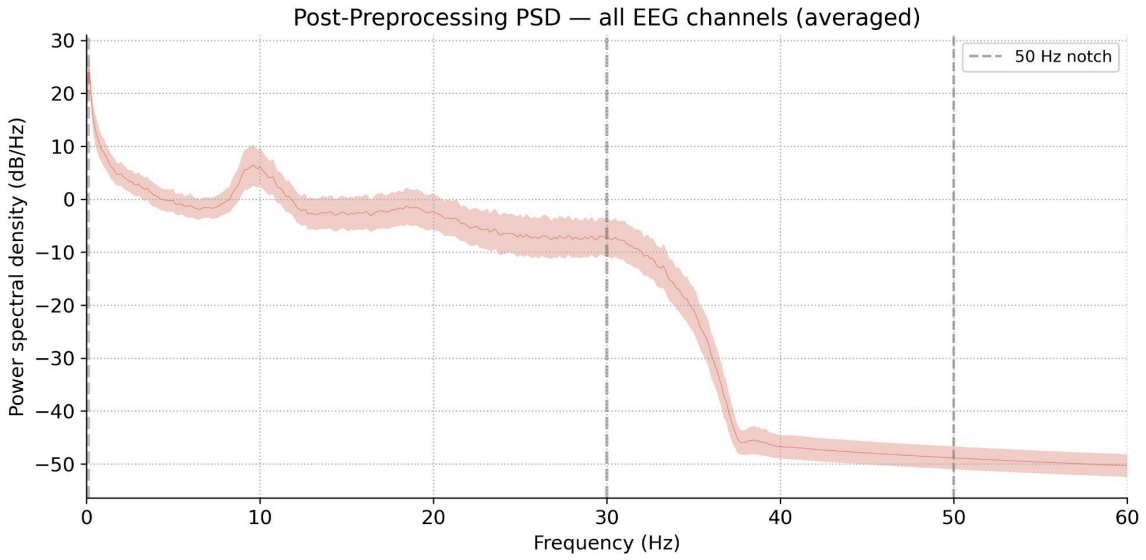


Figure-4: The Post-preprocessing PSD across all EEG channels wherein the 50 Hz peak is eliminated and the sharp rolloff above 30 Hz further confirms the bandpass filter application.

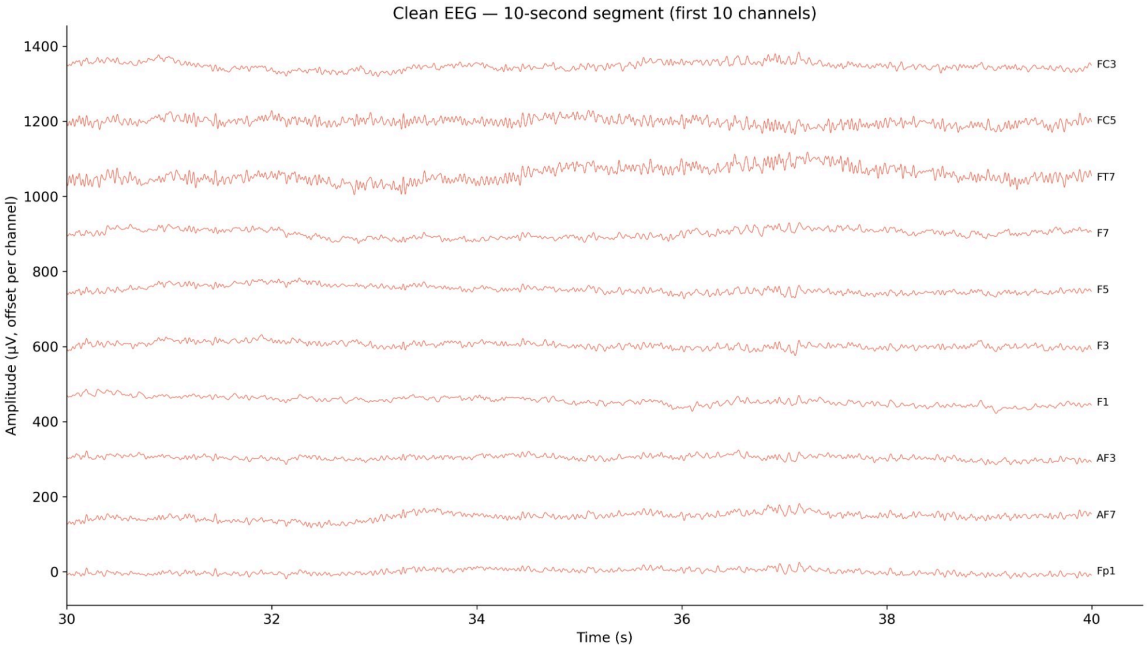


Figure-5: A cleaned EEG time-series (30-40s) for the first 10 channels indicating a reduced artifact contamination and clearer cortical oscillations.

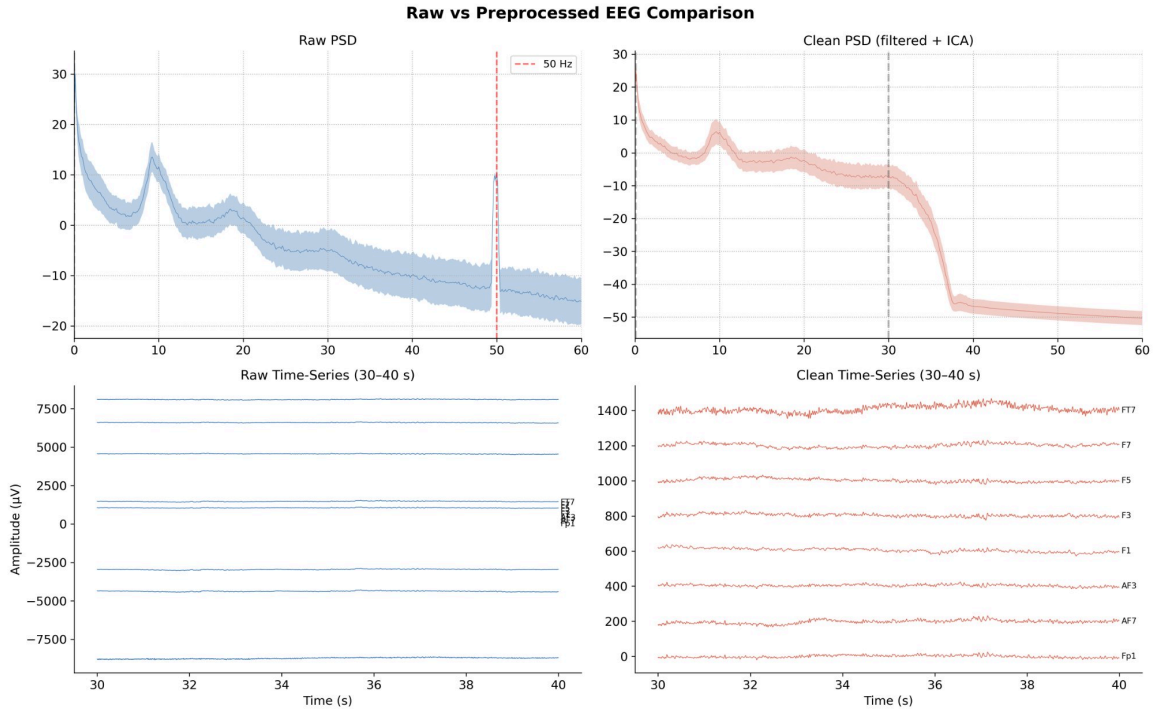


Figure-6: A side-by-side comparison of raw versus preprocessed EEG wherein the top panels highlight PSD before and after filtering/ICA whilst the bottom panels showcase the time-series before and after preprocessing.

II. Part-II: Epoching & Event-Related Potentials

A. Segmentation & Baseline Correction: The continuous EEG data were segmented into epochs from -200 ms to +800 ms relative to stimulus onset and the baseline correction was applied using the pre-stimulus interval (−200 to 0 ms), subtracting the mean voltage of each channel in that window from the entire epoch. Hence, an automatic artifact rejection threshold of 100 µV peak-to-peak was applied thereby removing epochs with residual large-amplitude deflections on any EEG channel.

B. Condition Averaging & Butterfly Plots: Notably, the surviving epochs were separated by condition (i.e., standard, oddball, noise) and were later averaged to exhibit evoked objects representing the ERP for each condition. The butterfly plots, with all channels overlaid and, Global Field Power or GFP, with the standard deviation across channels at each time point were thereby computed for each condition. Hence, the GFP here reveals time points of maximal scalp-wide activation.

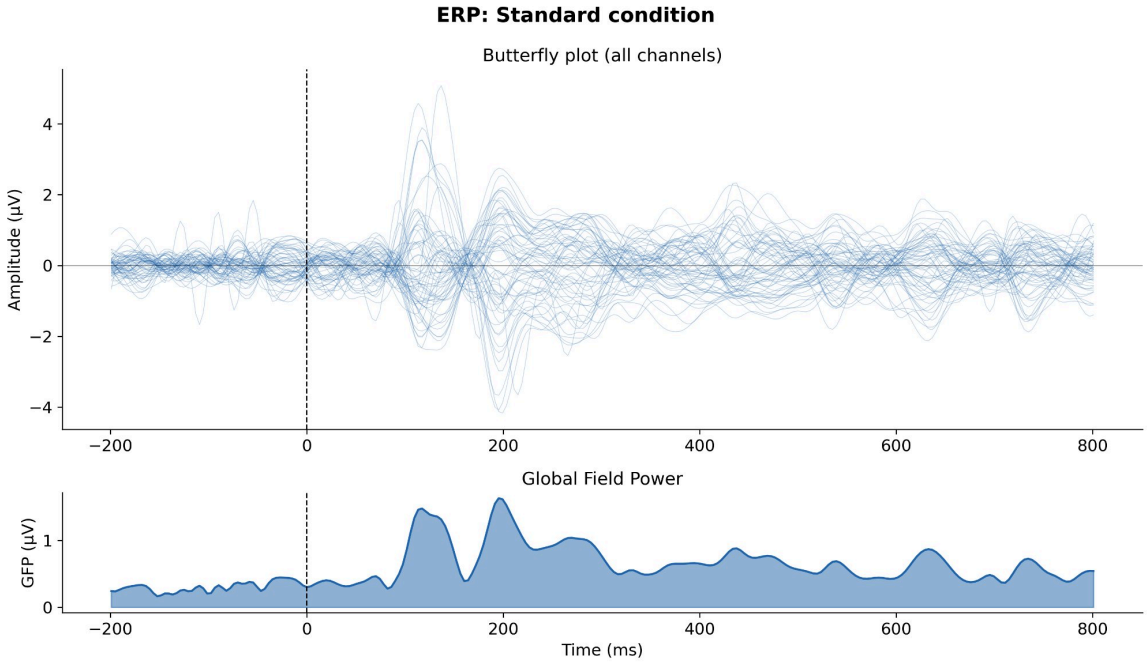


Figure-7: The butterfly plot and GFP for the standard condition here reveals that the activity is concentrated around early auditory components (i.e., N1/P1,~100-200 ms) with minimal late activity.

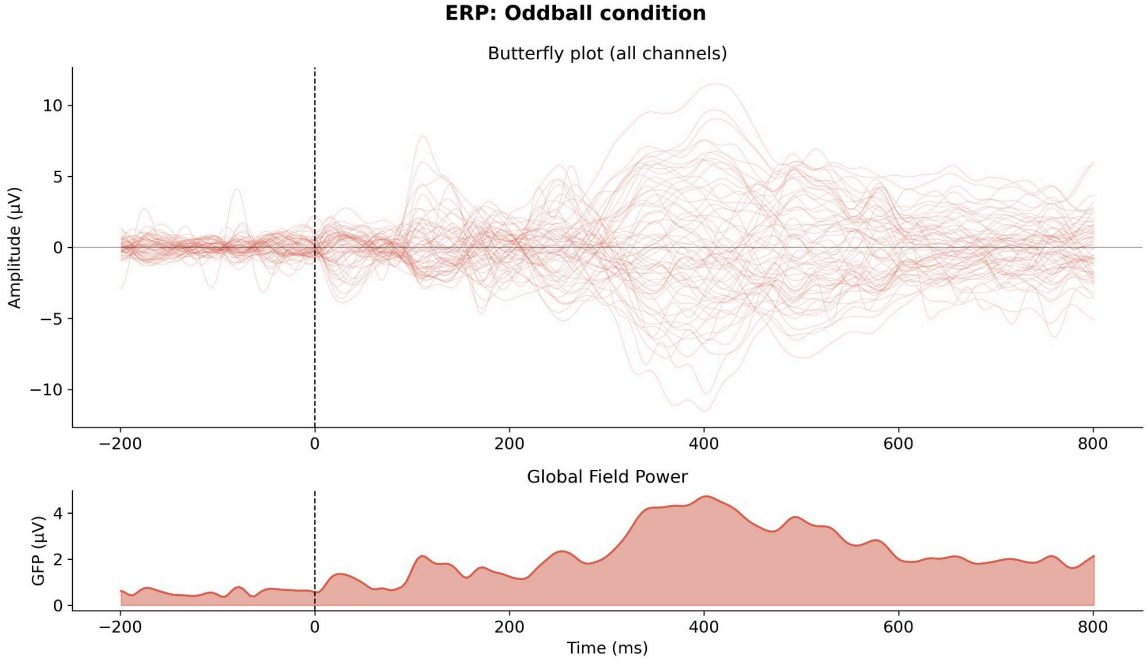


Figure-8: The butterfly plot and GFP for the oddball condition wherein a prominent late positive deflection (ie.e., at P300, ~300-500 ms) is visible across channels, the GFP peaking around 400 ms.

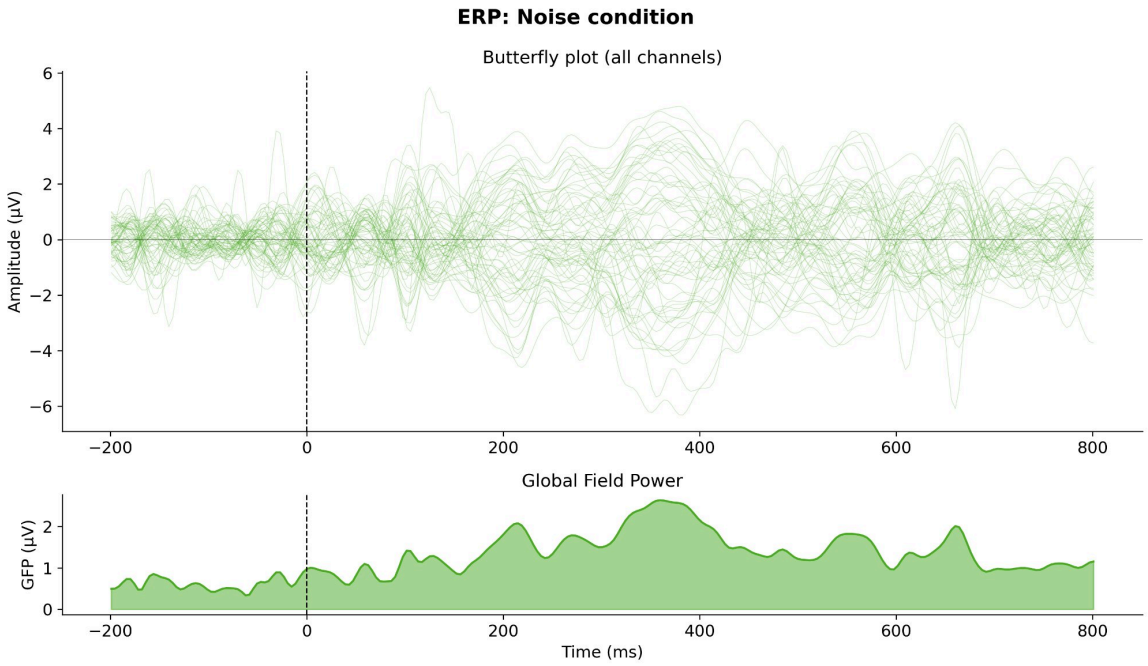


Figure-9: The Butterfly plot and GFP for the noise/distractor condition.

C. ERP Component 1 (P300):

RQ: Does the auditory oddball stimulus elicit a significantly larger P300 component compared to the standard stimulus present at parietal electrode Pz?

Hypotheses:
H^0: There is no significant difference in mean P300 amplitude (300–500 ms, Pz) between the standard and oddball conditions.
H^1: The oddball condition elicits a significantly larger (more positive) P300 amplitude at Pz compared to the standard condition.

Methodology: The P300 component was analyzed at three centro-parietal channels: a) Pz (primary region of interest), b) Cz and, c) CPz. The ERP waveforms for the standard and oddball conditions were plotted at these channels with a shaded region exhibiting the canonical P300 time-window at 300–500 ms. The P300 peak latency was identified as the time point of maximum positive amplitude in the oddball condition at Pz within the 300–500 ms window. Hence, the topographic maps were generated at this peak latency for both conditions. The single-trial mean amplitudes were extracted from the 300–500 ms time-window at Pz for each epoch and the Shapiro-Wilk test was applied to assess the normality of the amplitude distributions. Therefore, if normality was violated, a Mann-Whitney U test was at play or else, an independent-samples t-test was applied indicating one-tailed, oddball > standard. The effect size was computed as Cohen's d (for t-test) or rank-biserial correlation r (for Mann-Whitney U).

Results: The visual inspection of the ERP waveforms (Figure 10) revealed a clear positive deflection in the oddball condition peaking at 406.2 ms at Pz (peak amplitude = 6.69 µV), consistent with the expected P300 latency. The standard condition exhibited minimal amplitude in

this time-window. The topographic distribution (Figure 11) confirmed the expected centro-parietal maximum for the oddball P300. Single-trial mean amplitudes in the 300–500 ms window at Pz were extracted for statistical comparison. Shapiro-Wilk tests indicated that normality was violated for both the standard ($p < .001$) and oddball ($p = .026$) distributions, so a Mann-Whitney U test (one-tailed, oddball > standard) was used. The P300 mean amplitude at Pz was significantly larger for oddball stimuli ($M = 4.67$ µV, $SD = 8.11$, $n = 85$) than for standard stimuli ($M = 0.02$ µV, $SD = 7.71$, $n = 426$), $U = 23979.00$, $p < .001$, $r = .32$. The medium-to-large effect size is consistent with prior literature on the auditory oddball paradigm (Polich, 2007).

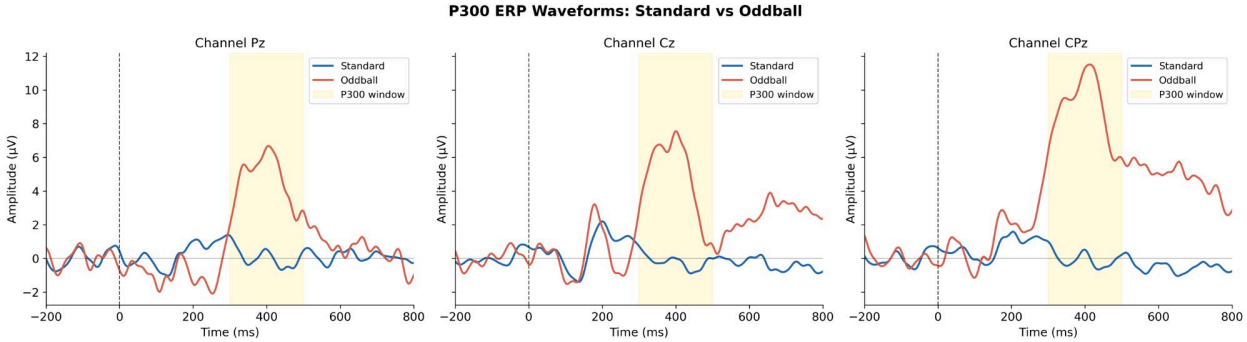


Figure-10: The P300 waveforms at Pz, Cz, and CPz for standard (blue) and oddball (red) conditions are indicated here wherein the shaded yellow region accounts the 300-500 ms and P300 time-window. The oddball condition here indicates a clear positive deflection that peaks around 406 ms.

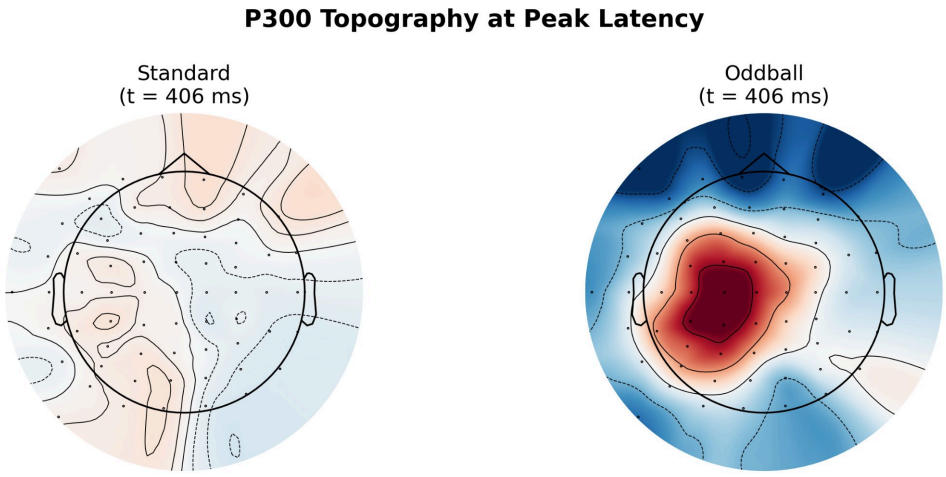


Figure-11: The topographic distribution at P300 peak latency (406 ms) wherein the standard condition indicates minimal activity whilst the oddball condition exhibits a strong centro-parietal positivity which is furthermore consistent with the P300 generator.

Discussion: The finding of a significantly enhanced P300 for oddball stimuli is consistent with the context-updating theory of P300 generation (Donchin & Coles, 1988; Polich, 2007). According to this framework, the P300 exhibits working-memory updating i.e., when an event deviates from the current mental model of the stimulus stream the attentional resources are directed toward revising that model itself further generating the centro-parietal positive potential. The rarity of the oddball (~15% probability) relative to the standard (~70%) indicates that each oddball perhaps carries greater surprise or information content, since the P300 amplitude scales inversely with stimulus probability and directly with task relevance (Polich, 2007). The centro-parietal distribution, maximal at Pz, indicates contributions from hippocampal and parieto-frontal cortical generators

involved in memory encoding and attention. Hence, the relatively long latency (~400 ms) generally indicates the higher-order cognitive processing as opposed to the primary sensory registration.

D. ERP Component 2 (N200):

RQ: Does the auditory oddball stimulus elicit a significantly larger N200 component as opposed to the standard stimulus at frontocentral electrode Fz?

Hypotheses:

H^0: There is no significant difference in N200 mean amplitude (180–280 ms, Fz) between the standard and oddball conditions.

H^1: The oddball condition elicits a significantly larger (more negative) N200 amplitude at Fz compared to the standard condition.

Methodology: The N200 component was analyzed at three frontocentral channels: a) Fz (primary ROI), b) Cz and, c) FCz. The ERP waveforms were subsequently plotted with inverted y-axis (i.e., the negative up, following ERP convention) and a shaded region exhibiting the N200 time-window at 180–280 ms. The N200 peak latency was identified as the time-point of maximum negative amplitude in the oddball condition at Fz within the window, therefore, the topographic maps were generated at this peak latency. Here, the single-trial mean amplitudes were extracted from the 180–280 ms window at Fz. In addition, the same normality assessment and statistical testing approach as the P300 analysis was applied (i.e., one-tailed, oddball being more negative than standard).

Results: The visual inspection (Figure 12) revealed a negative deflection in the oddball condition peaking at 253.9 ms at Fz (peak amplitude = −3.96 µV), consistent with the expected N200 latency. The topographic distribution (Figure 13) exhibited frontocentral negativity for the oddball condition, characteristic of the N200 generator in auditory cortex. Single-trial mean amplitudes in the 180–280 ms window at Fz were extracted for statistical comparison. Shapiro-Wilk tests indicated that the assumption of normality was met for both the standard (p = .202) and oddball (p = .675) distributions, so an independent-samples t-test (one-tailed, oddball more negative than standard) was used. The N200 mean amplitude at Fz was significantly more negative for oddball stimuli (M = −1.46 µV, SD = 4.90, n = 85) than for standard stimuli (M = 1.32 µV, SD = 4.62, n = 426), $t(509) = 5.00$, $p < .001$, $d = 0.60$. The medium effect size supports the hypothesis that deviant auditory events elicit enhanced pre-attentive mismatch processing.

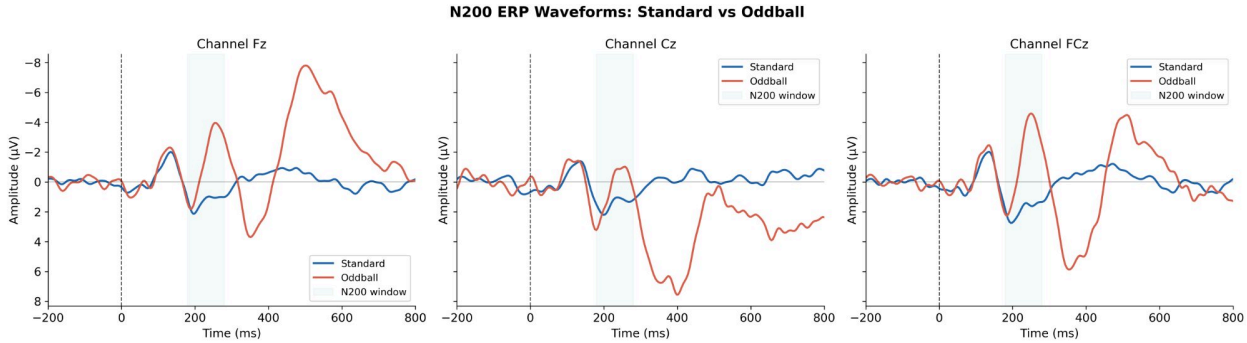


Figure-12: The N200 ERP waveforms at Fz, Cz, and FCz for the standard (i.e., blue) and the oddball (i.e., red) conditions wherein the shaded blue region accounts the 180-280 ms (N200) window. Note the inverted y-axis denoting the negative up.

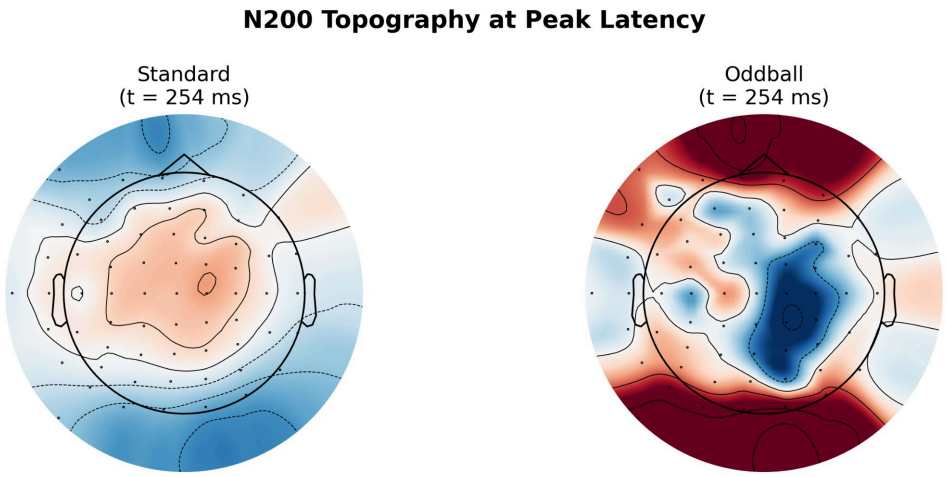


Figure-13: The topographic distribution at N200 peak latency (254 ms) wherein the oddball condition exhibits a frontocentral negativity which is consistent with the N200 generator in auditory cortex.

Discussion: The N200 component shows pre-attentive deviance detection and automatic discrimination of a stimulus as different from the prevailing auditory template (Näätänen et al., 2007). Here, notably two interpretations are relevant: a) that the frontocentral N200 is closely related to the Mismatch Negativity (MMN) which was generated in auditory cortex and supplementary motor areas, since it reflects automatic change-detection that operates without directed attention (Näätänen et al., 2007), and; b) since the task required a button-press to oddball targets, here the N200 also engages an N2b subcomponent (~200–300 ms) exhibiting attentional orienting to the deviant as followed by a P3a novelty response and the canonical P3b at centro-parietal sites. Therefore, the combination of enhanced N200 and P300 to oddball stimuli reveals the full hierarchy of auditory deviance processing i.e., from early automatic change-detection (N200/ MMN) via controlled attentional updating and working-memory revision (P300) (Polich, 2007; Näätänen et al., 2007).

E. Mismatch Negativity (MMN) Difference Wave: The MMN was isolated by computing the difference wave (i.e., oddball evoked minus standard evoked). This subtraction essentially cancels the common auditory response and further reveals the neural signature of automatic change detection. The MMN typically peaks between 100–250 ms over frontocentral scalp and inverts polarity at the mastoids, representing generators in auditory cortex (Näätänen et al., 2007). The difference wave at Fz (Figure 15) exhibited a clear negative deflection peaking at approximately 250 ms with an amplitude of −4.92 μV, consistent with the MMN. This component indicates that the activation of a pre-attentive auditory change-detector i.e., the auditory cortex maintains a predictive model of the recent stimulus sequence, and even when the oddball violates this prediction, a prediction-error signal manifests as the MMN itself. Therefore, this interpretation aligns well with the predictive coding models of perception where in the brain continuously generates predictions about upcoming sensory input, and discrepancies further generate error signals that propagate up the cortical hierarchy (Näätänen et al., 2007).

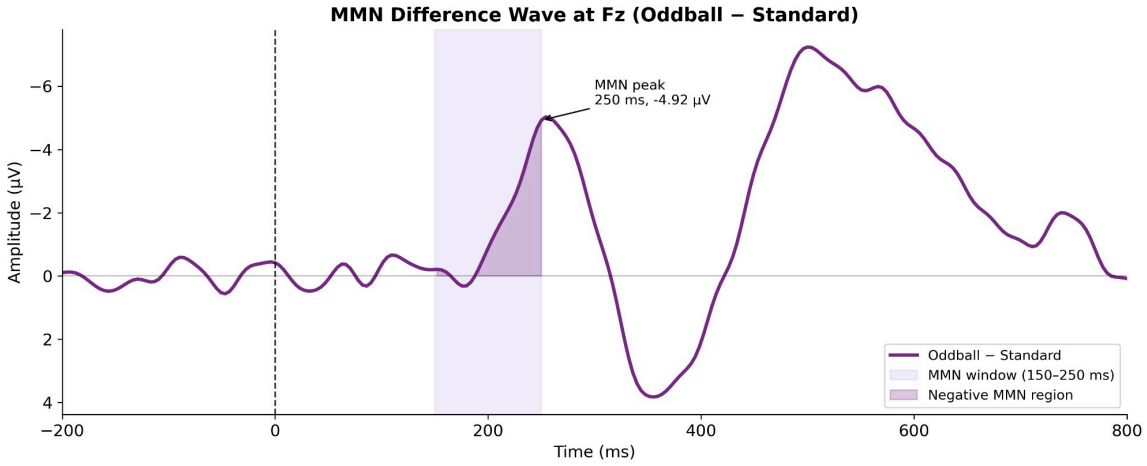


Figure-14: The MMN difference wave (i.e., oddball minus standard) at Fz wherein the negative deflection peaking at approximately 250 ms (−4.92 µV) is consistent with the MMN. Here, the purple shaded region exhibits the negative MMN region within the 150–250 ms analysis window.

III. Part-III: Reference Electrode Comparison

A. Method: The preprocessed data (i.e., prior to average re-referencing) were re-referenced via four different schemes: (a) the average reference with the mean of all EEG channels, (b) the Cz reference with the vertex electrode, and (c) the linked mastoids (i.e., TP7+TP8, with the nearest available channels to M1/M2 in this BioSemi montage), and (d) Fpz reference with the nasion proxy. Notably, for each scheme, epochs were extracted and averaged for the oddball condition, whilst a topographic map was generated at the P300 peak latency (406 ms).

B. Results:

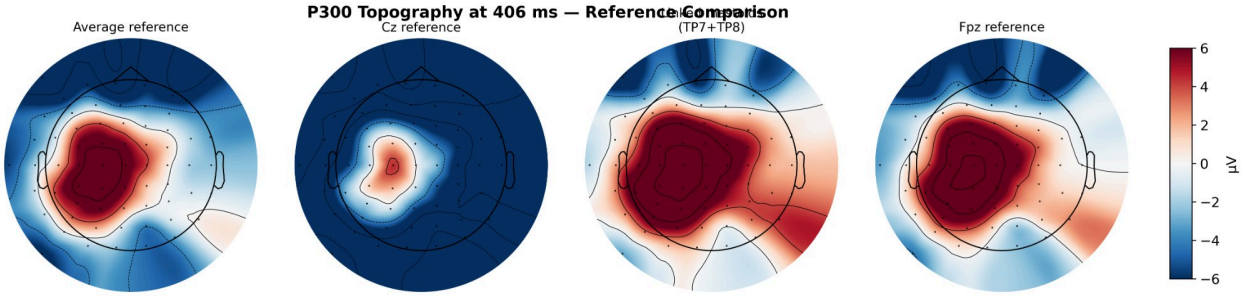


Figure-15: The P300 topography at 406 ms under four referencing schemes wherein the average reference exhibits a balanced centro-parietal positivity whilst the Cz reference suppresses the P300 due to Cz being active during the response. Additionally, the linked mastoids enhance parietal positivity, whilst the Fpz reference introduces frontal distortion.

C. Discussion: Essentially, every EEG recording measures a potential difference present between an active electrode and a reference. If the true scalp potential at electrode i is φi and the reference is φr, then the resultant recorded signal is Vi = φi − φr. Here, changing the reference is equivalent to subtracting a constant offset from all channels and while no reference is neutral, here each choice introduces a bias toward minimizing apparent activity at the reference location. So, the average reference itself produced the most balanced topography, with the P300 positivity clearly isolated over centro-parietal scalp. This happens because the average reference

basically approximates a true reference-free distribution when electrode coverage is uniform in nature (Dien, 1998). Here, the Cz reference suppressed the apparent P300 at Pz because Cz itself is active during the P300 response, hence, subtracting an active reference from neighboring channels just reduces their apparent amplitude. In addition, the linked mastoid reference (i.e., TP7+TP8) significantly enhanced the parietal positivity and further produced good signal-to-noise for auditory ERPs, but introduced a bilateral temporal bias. Notably, the Fpz reference caused severe frontal distortion by artificially spreading the P300 positivity anteriorly. Therefore, for mapping occipito-temporal activity, the linked mastoid reference gives a useful alternative since the mastoid reference sites are relatively distant from the generators, but the average reference remains optimal for this 64-channel dataset due to its uniform coverage and minimal regional bias (Dien, 1998; Yao, 2001).

References

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